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DDAR Exercise 4.2

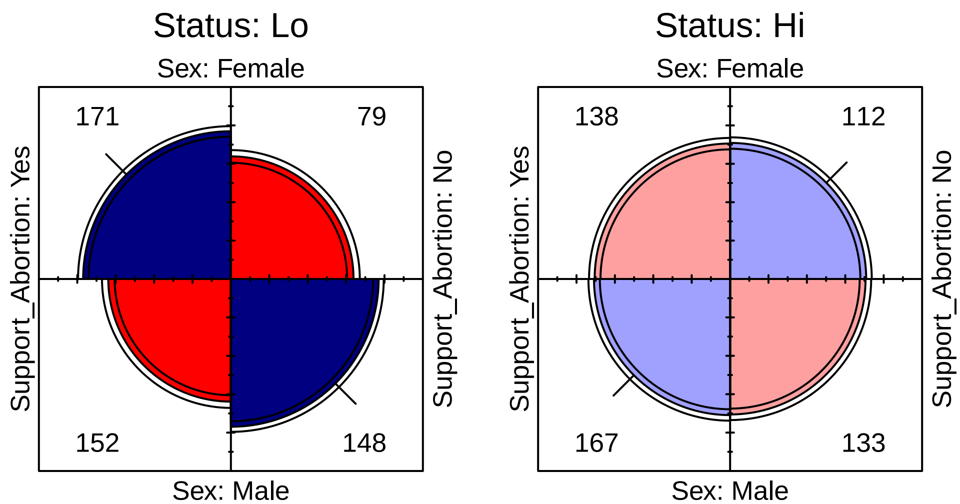
The data set Abortion in vcdExtra gives a $2 \times 2 \times 2$ table of opinions regarding abortion in relation to sex and status of the respondent. This table has the following structure:

```
data("Abortion", package = "vcdExtra")
str(Abortion)
```

```
## 'table' num [1:2, 1:2, 1:2] 171 152 138 167 79 148 112 133
## - attr(*, "dimnames")=List of 3
##  ..$ Sex             : chr [1:2] "Female" "Male"
##  ..$ Status          : chr [1:2] "Lo" "Hi"
##  ..$ Support_Abortion: chr [1:2] "Yes" "No"
```

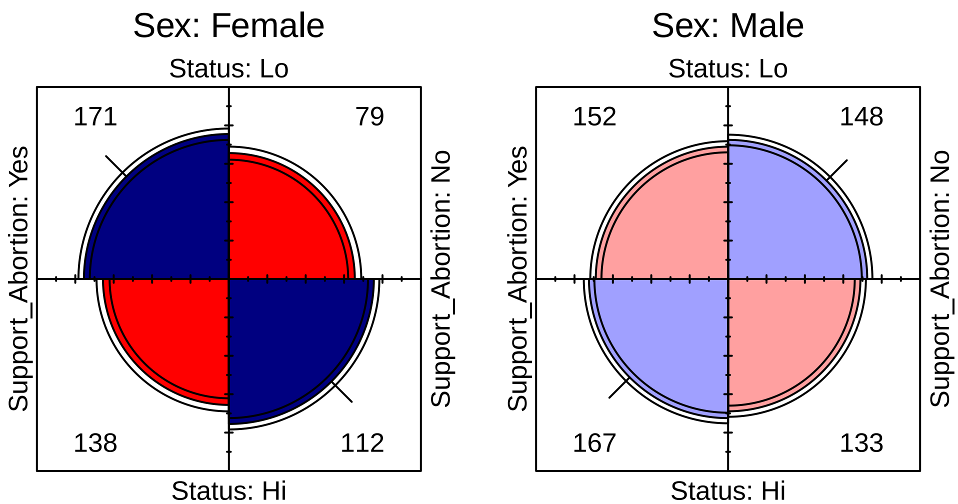
(a) *Taking support for abortion as the outcome variable, produce fourfold displays showing the association with sex, stratified by status.*

```
strat_status <- aperm(Abortion, c(1, 3, 2))
fourfold(strat_status)
```



(b) *Do the same for the association of support for abortion with status, stratified by sex.*

```
strat_sex <- aperm(Abortion, c(2, 3, 1))
fourfold(strat_sex)
```



(c) *For each of the problems above, use* ***`oddsratio()`*** *to calculate the numerical values of the odds ratio, as stratified in the question.*

```
oddsratio(strat_status, log = FALSE) # 4.2a
```

```
##  odds ratios for Sex and Support_Abortion by Status
##
##        Lo        Hi
## 2.1075949 0.9812874
```

```
oddsratio(strat_sex, log = FALSE)     # 4.2b
```

```
##  odds ratios for Status and Support_Abortion by Sex
##
##    Female      Male
## 1.7567419 0.8179317
```

(d) *Write a brief summary of how support for abortion depends on sex and status.*

In the fourfold plots stratified by status, when looking at low status individuals, we see that support for abortion depended on one's gender, with odds ratio significantly greater than one. Moreover, low status individuals were more likely to support abortion, whereas high status individuals were more likely to be against abortion. In contrast, support for abortion did not depend on sex when individuals were of high status, with odds ratio not significantly different from one.

In the fourfold plots stratified by gender, when looking at females, support for abortion depended on one's status, with odds ratio significantly greater than one. Moreover, low status females were more likely to support abortion, whereas high status females were more likely to be against abortion. As for males, support for abortion did not depend on status, with odds ratio not significantly different from one.

DDAR Exercise 4.4

The `Hospital` *data in vcd gives a 3 × 3 table relating the length of stay (in years) of 132 long-term schizophrenic patients in two London mental hospitals with the frequency of visits by family and friends.*

(a) *Carry out a χ^2 test for association between the two variables*

```
data("Hospital")
chisq.test(Hospital)
```

```
##
##  Pearson's Chi-squared test
##
## data:  Hospital
## X-squared = 35.171, df = 4, p-value = 4.284e-07
```

(b) *Use assocstats() to compute association statistics. How would you describe the strength of association here?*

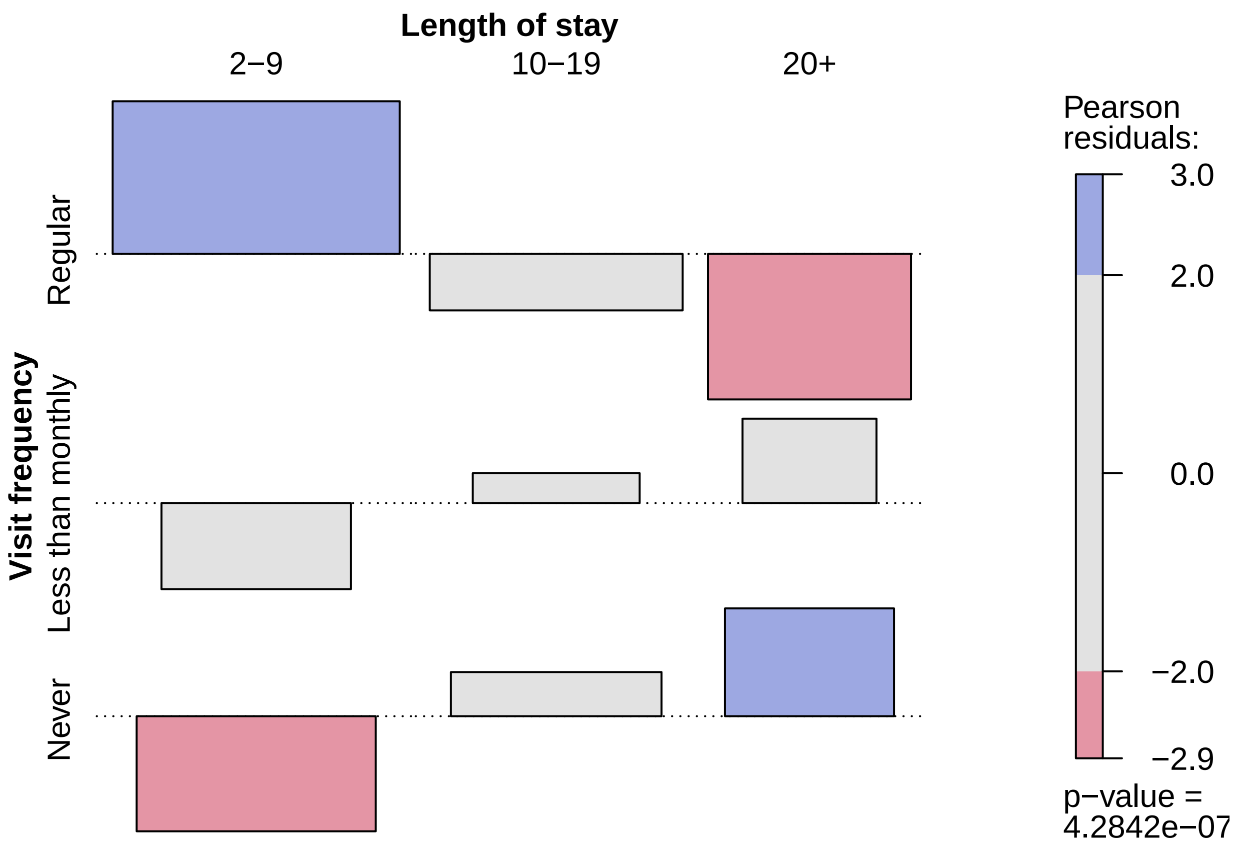
```
(hosp_stat <- assocstats(Hospital))
```

```
##                     X^2 df   P(> X^2)
## Likelihood Ratio 38.353  4 9.4755e-08
## Pearson          35.171  4 4.2842e-07
##
## Phi-Coefficient   : NA
## Contingency Coeff.: 0.459
## Cramer's V        : 0.365
```

The association was significant ($\chi^2(4) = 35.171$, $p < .001$), with Cramer's $V = 0.365$ indicating the strength of association was substantial.

(c) *Produce an association plot for these data, with visit frequency as the vertical variable. Describe the pattern of the relation you see here.*

```
assoc(Hospital, shade = TRUE)
```



The association plot depicts that those with more frequent visits to the hospital have a lower length of stay than would be expected under independence. Conversely, those with less frequent visits to the hospital have a greater length of stay than would be expected under independence.

(d) *Both variables can be considered ordinal, so `CMHtest()` may be useful here. Carry out that analysis. Do any of the tests lead to different conclusions?*

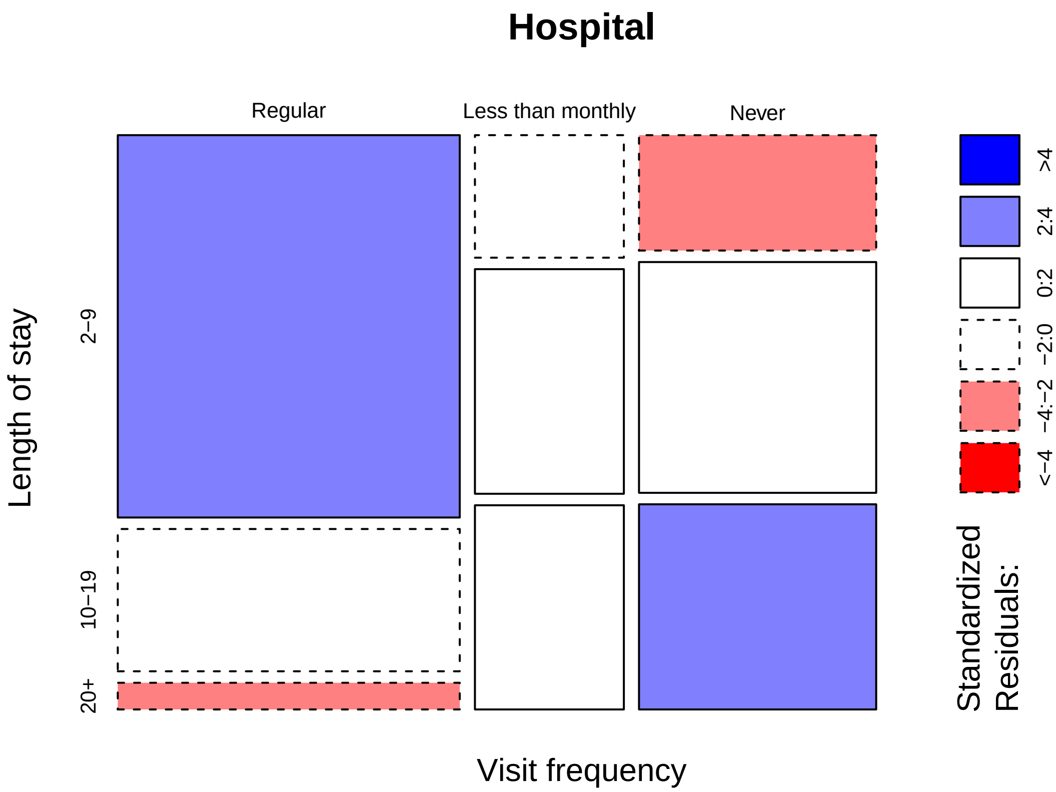
```
CMHtest(Hospital)
```

```
## Cochran-Mantel-Haenszel Statistics for Visit frequency by Length of stay
##
##                  AltHypothesis  Chisq Df       Prob
## cor        Nonzero correlation 29.138  1 6.7393e-08
## rmeans  Row mean scores differ 34.391  2 3.4044e-08
## cmeans  Col mean scores differ 29.607  2 3.7233e-07
## general    General association 34.905  4 4.8596e-07
```

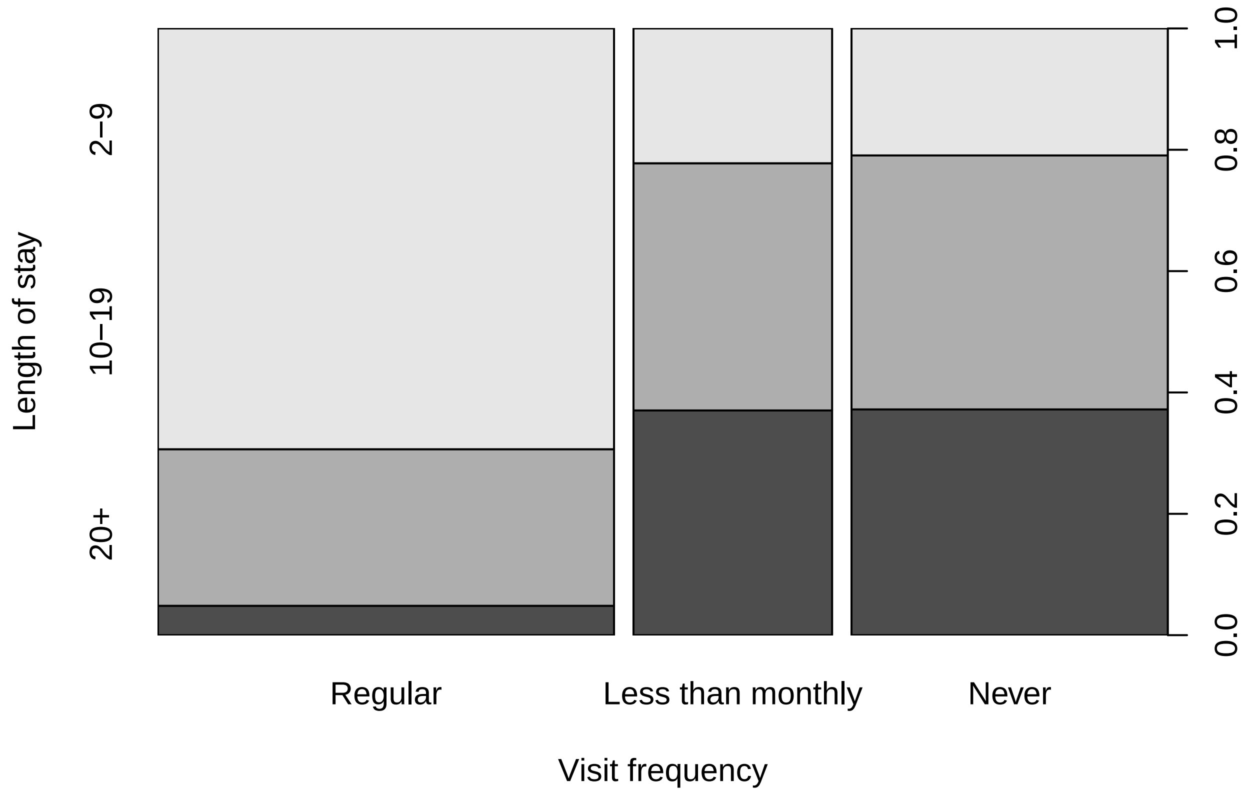
No, the tests all lead to the conclusion that there is a significant departure from independence in the visit frequency and length of stay variables.

(e) *Try one or more of the following other functions for visualizing two-way contingency tables with this data: `plot()`, `tile()`, `mosaic()`, and `spineplot()`. [For all except `spineplot()`, it is useful to include the argument `shade=TRUE`].*

```r
plot(Hospital, shade = TRUE)
```




```r
spineplot(Hospital, shade = TRUE)
```



(f) *Comment on the differences among these displays for understanding the relation between visits and length of stay.*

These displays show the same pattern of association depicted in question 4.4c. The mosaic display is easier to read at first glance because of the shading.

DDAR Exercise 4.6

The two-way table `Mammograms` *in vcdExtra gives ratings on the severity of diagnosis of 110 mammograms by two raters.*

(a) *Assess the strength of agreement between the raters using Cohen's κ, both unweighted and weighted.*

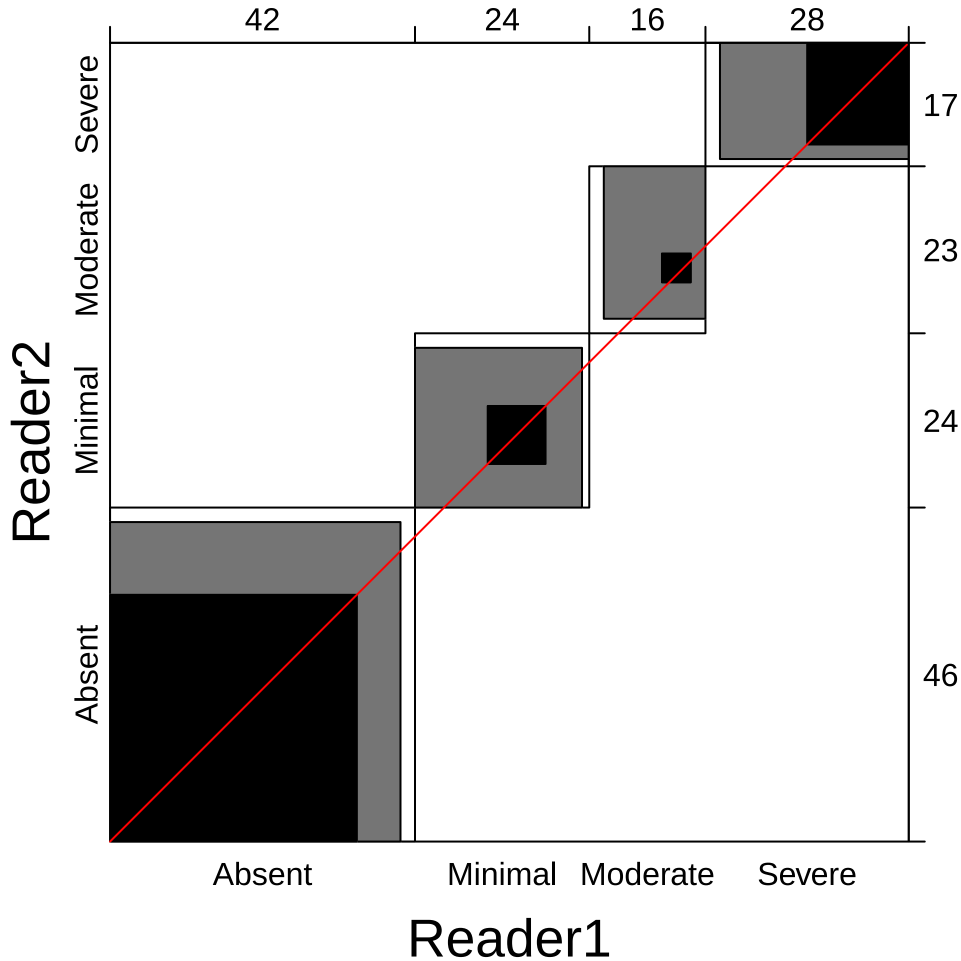
```
data("Mammograms")
(kaps <- Kappa(Mammograms))
```

```
##              value     ASE      z
## Unweighted 0.3713 0.06033  6.154
## Weighted   0.5964 0.04923 12.114
```

Since κ ranges from 0 to 1, both the unweighted $\kappa = 0.371$ and the weighted $\kappa = 0.5963693$ indicate moderate to moderately strong agreement between raters.

(b) *Use* `agreementplot()` *for a graphical display of agreement here.*

```r
agreementplot(Mammograms)
```



(c) *Compare the Kappa measures with the results from* `assocstats()`*. What is a reasonable interpretation of each of these measures?*

```r
assocstats(Mammograms)
```

```
##                     X^2 df   P(> X^2)
## Likelihood Ratio 92.619  9 4.4409e-16
## Pearson          83.516  9 3.2307e-14
##
## Phi-Coefficient   : NA
## Contingency Coeff.: 0.657
## Cramer's V        : 0.503
```

The results from `assocstats()` indicate a significant departure from independence, with Cramer's V indicating a strong association between raters. This makes sense, as the Kappa measures indicate substantial agreement (and thus, association) between raters.